Dylan Petzko

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SUMMARY

Full Stack and DevSecOps Engineer with 5+ years of experience building mission-critical software for DOD programs at Lockheed Martin. Skilled in scalable web application development, RESTful API design, system integrations, and CI/CD pipelines across cloud and air-gapped environments. Experienced in React-based frontend development, Python/Flask backends, LLM-powered tooling, and AWS infrastructure — with a consistent record of reducing operational burden through automation and well-architected software.

WORK EXPERIENCE

Lockheed Martin, RMS — Mount Laurel, NJ

Aug 2022 – Current

Full Stack Engineer II | Cloud Current Role — Dec 2025 – Current

- Engineered and managed production-grade Kubernetes clusters using Helm, streamlining deployment of Keycloak and Kafka across restricted, air-gapped environments.
- Maintained and extended AWS cloud infrastructure to support auxiliary services, applying Terraform for infrastructure-as-code provisioning and configuration management.
- Architected monitoring stack using Prometheus, Grafana, and Loki for real-time visibility and log aggregation across mission-critical DOD infrastructure.
- Implemented STIG compliance and navigated security requirements to ensure all deployments meet the highest DOD security standards.

Software Engineer II | AI Data Analysis Concurrent with Cloud — Dec 2025 – Current

- Serving as software architect on an AI-assisted data-analysis platform leveraging large language models to build agentic workflows that enable engineers to rapidly discover and process large volumes of data.
- Designed and developed scalable full-stack web application using Flask RESTful APIs (backend) and React with component-based architecture (frontend), with Tauri providing a native cross-platform interface.
- Architected system integrations between the LLM agent layer, data sources, and frontend to enable seamless data sharing across application boundaries.
- Containerized application using Docker and authored Helm charts for deployment as a standalone web service and as a Kubernetes-native pod.
- Led requirements gathering, system modeling, and architectural design documentation for the platform.

Software Engineer II | Test Automation Concurrent with DevSecOps — Oct 2023 – Dec 2025

- Developed and maintained proprietary automated testing framework for end-to-end regression testing using Python, Jenkins, and GitLab CI/CD — estimated cost-savings of over $500,000/year.
- Collaborated with customer program leadership to drive development priorities and improve product reliability; created quality and performance metric documentation.
- Architected automated testing tools in Python, saving $200,000 in development costs year-over-year.

Software Engineer | DevSecOps Concurrent with Test Automation — Feb 2023 – Dec 2025

- Built inner-source Python data-analysis library for proprietary formats, reducing analysis time by 75% and saving over $750,000 in labor costs per year.
- Created end-to-end CI/CD pipelines using Docker, Vagrant, Splunk, Ansible, and GitLab.
- Developed automated deployment tool for container, virtual, and baremetal environments with estimated savings of $1,000,000 year-over-year.

Systems Engineer | Integration and Test — Aug 2022 – Oct 2023

- Worked on-site with foreign military customers to troubleshoot and update software builds.
- Coordinated, developed, and executed detailed test procedures for software products.
- Built post-test data analysis tool for automated reporting using Python, Matlab, and Tableau.
- Developed and modified software requirement documentation using Jazz and DOORS.

Stilog I.S.T — New York, NY

Jan 2021 – Aug 2022

- Delivered tailored software solutions for customers using a proprietary framework; utilized Java, HTML, and CSS to meet requirements.
- Created comprehensive business requirement documentation to translate customer needs into actionable design plans.
- Maintained and managed customer product deployments on AWS and Azure, including infrastructure provisioning and environment upkeep.

SKILLS

Languages	Python, Bash, Go, Groovy/Java, Rust, C++, SQL
OS Platforms	UNIX (Solaris 8/9/10), RHEL (6–9), Ubuntu, NixOS
Provisioning	Terraform, AWS, ARM, Kubernetes, Docker, MiniKube
Virtualization	VMWare, Vagrant, KVM
Web & Native	React, Flask, Tauri
CI Tools	Jenkins/Hudson, GitLab, ArgoCD, FluxCD
Config Management	Ansible, Nix, Nginx, Chef, Puppet, Maven, Make
Monitoring	Prometheus, Grafana, Loki, Splunk
Messaging & Auth	Kafka, Keycloak
AI & ML	LLM Integration, Agentic Workflows, AI-assisted Data Analysis

EDUCATION

Rowan University	B.S. Computing and Informatics, Minor in CyberSecurity	2016 – 2020
Rowan University	B.S. Emergency Management and Homeland Security, Minor in Counter-Terrorism	2016 – 2020